Quantum Computing with the Exchange Interaction




spin quantum computers without single spin control



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work done in collaboration with:

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Cult of Universality



Alan Turing

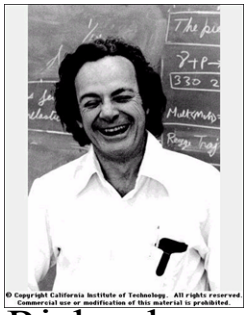
Turing Machine: basic computing machine that quantifies our notion of *"carrying out an algorithmic task"*

Universal Turing Machine: Turing machine which can be used to *efficiently simulate* any Turing machine.

but…. "Information is physical"

+ "Nature is quantum mechanical" =

Theory of Computation based on Quantum Mechanics



Richard Feynman

Universal Quantum Computer: physical device which can be used to efficiently simulate any "algorithmic task" carried out on information presented in a quantum system

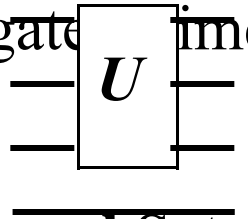
(Feynman, Deutsch, Benioff, Barenco, Ekert, Yao, DiVincenzo, Lloyd, Kitaev, Solovay)

Universal Sets of Quantum Gates

Quantum Circuit Model

wire = carrier of quantum information (qubit $\{|0\rangle, |1\rangle\}$, qudit $\{|0\rangle, |1\rangle, \ldots, |d\rangle\}$)

gate = time evolution of the quantum information



$$\boldsymbol{U} = \mathrm{T}\left[\exp\left(-i\boldsymbol{H}(t)t_0/\eta\right)\right]$$

Universal Sets of Quantum Gates

A set of gates G is a universal set of gates if

- each $\boldsymbol{U}i \in G$ acts on a bounded number of qubits (locality)
- any quantum circuit (performing any desired unitary operation) can be efficiently *simulated* to arbitrary accuracy by a circuit with gates in G

(arbitrary accuracy: arbitrary distance$(\boldsymbol{U},\boldsymbol{V}) = \max|\psi\rangle||(\boldsymbol{U}-\boldsymbol{V})|\psi\rangle||$)

Information is Fungible

Fungible *adj.* (New Latin *fungibilis*, from Latin *fungi* to perform)
1. being of such a nature that one part or quantity may be replaced by another equal part or quantity in the satisfaction of an obligation.
2. interchangable.

Value of classical information is independent of the medium:
- spin of millions of electrons
- presence or absence of a chad[1]
- etc.

Classical information is fungible. (can be encoded)

Quantum information is fungible. (can be encoded)

[1] U.S. Supreme Court case no. 00-949, George W. Bush vs. Albert Gore Jr.

The Stereotypical Quantum Comptuer

1. Physical identification of qubits (spin, hyperfine levels, etc.)

2. Interactions for single qubit gates

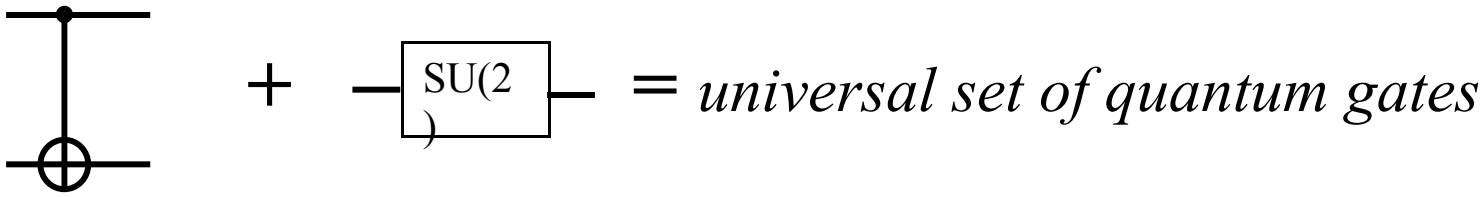
$$U(\theta, \hat{n}, \phi) = e^{i\phi}\left(\cos\theta\, \mathbf{I} + i \sin\theta\, \hat{n}\cdot\vec{\sigma}\right)$$

(phase ϕ is not relevant)

3. Interactions for a non-trivial two-body gate (controlled-not gate, controlled-phase gate, etc.)

$|0\rangle|0\rangle \to |0\rangle|0\rangle$, $|0\rangle|1\rangle \to |0\rangle|1\rangle$,

$|1\rangle|0\rangle \to |1\rangle|1\rangle$, $|1\rangle|1\rangle \to |1\rangle|0\rangle$, controlled-not



Encoded Universality

Physical System ☐ single qubit gates + CNOT

independent of systems natural talents (fast, robust interactions)

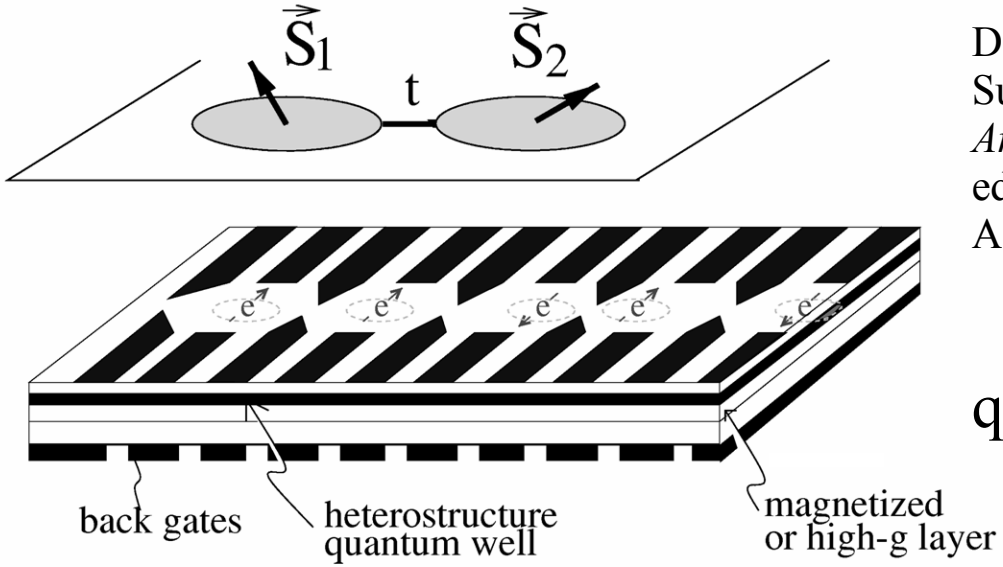
only need to simulate a quantum circuit ☐ encode the quantum information

many interactions between qubits are not "fully universal" but can be made universal with a little bit of encoding: "encoded universal"

~~"fully universal": unit space efficiency - gates acting on *n* qubits can~~ be used to implement any unitary gate on *n* qubits

"encoded universal": non-unit space efficiency - gates acting on *n* qubits can be used to simulate unitary gates on *En* qubits (*E=encoding efficiency*)

Spintronic Quantum Computer



D.P. DiVincenzo, G. Burkard, D. Loss, E.V. Sukhorukov, in *Quantum Mesoscopic Phenomena And Mesoscopic Devices In Microelectronics,* eds. I.O. Kulik and R. Elliatioglu (NATO Advanced Study Institute, Turkey) 1999

qubit = spin of single electron

exchange interaction between qubits used for two-qubit interaction: tuned by raising and lowering barrier between dots to change overlap.

coupling strength: on ≈0.1 meV, off many orders of magnitude smaller

~~single qubit gates need to supplement exchange to make "fully~~ universal"

single qubit gates = huge demands on g-factor engineering, strong inhomogeneous magnetic fields, slow microwave manipulations.

Exchange Alone Universal?

Exchange (Heisenberg) coupling between qubits i,j: $\mathbf{E}_{i,j} = \sigma_i \cdot \sigma_j$

three qubits:

$$\mathbf{H}_0 = \mathbf{E}_{1,2} + \mathbf{E}_{2,3} + \mathbf{E}_{1,3},$$

$$\mathbf{H}_1 = \tfrac{1}{4\sqrt{3}}(\mathbf{E}_{1,3} - \mathbf{E}_{2,3}),\ \mathbf{H}_3 = \tfrac{1}{12}(-2\mathbf{E}_{1,2} + \mathbf{E}_{2,3} + \mathbf{E}_{1,3}),\ \mathbf{H}_2 = i[\mathbf{H}_1, \mathbf{H}_3]$$

such that

$$[\mathbf{H}_0, \mathbf{H}_\mu] = 0, \quad [\mathbf{H}_i, \mathbf{H}_j] = i\varepsilon_{ijk}\mathbf{H}_k$$

su(2)!

Abelian invariant subalgebra = irrelevant phase

Question: Given a set of interaction Hamiltonians **H**i, what operations can be performed by turning on and off these interactions?

Answer: *interactions in the Lie algebra generated by the **H**i* (Lloyd, DiVincenzo, Deutsch et. al.)

Encoded Action

4 fold degenerate, 1 dimensional irreducible representation of *su(2)*

$$\mathbf{H}_i = \left[\begin{array}{ccc|c} 0 & | & 0 & \\ \cdots & \cdots & \cdots & 0 \\ 0 & | & 0 & \\ \hline & & & \sigma_i \;\vdots\; 0 \\ & 0 & & \cdots\cdots\cdots \\ & & & 0 \;\vdots\; \sigma_i \end{array}\right]$$

2 fold degenerate, 2 dimensional irreducible representation of *su(2)*

Encode one qubit of information into three physical qubits:

$$|0_L\rangle = \tfrac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right)|0\rangle$$
$$|1_L\rangle = \sqrt{\tfrac{2}{3}}|00\rangle|1\rangle - \sqrt{\tfrac{1}{3}}\left\{\tfrac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right)\right\}|0\rangle$$

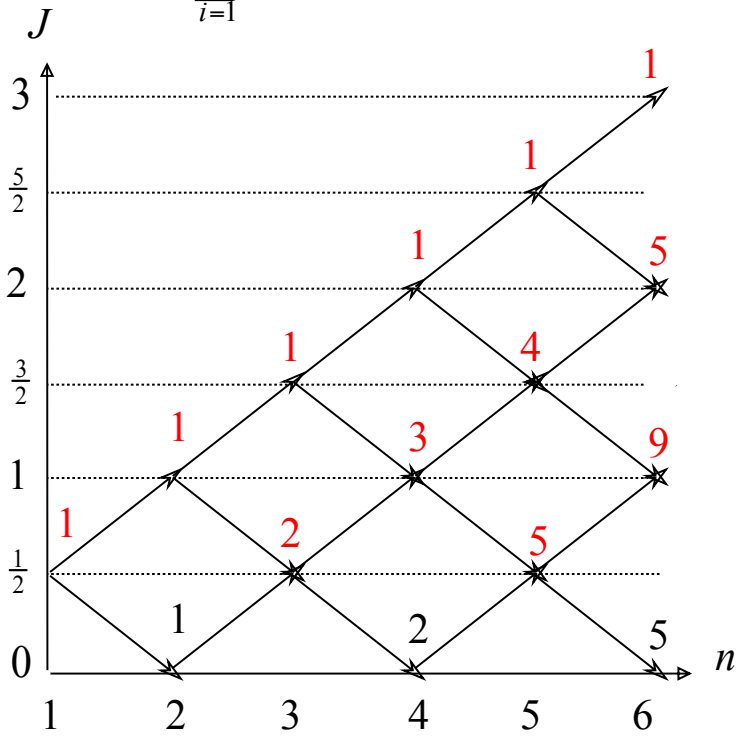
or

$$|0_L\rangle = \tfrac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right)|1\rangle$$
$$|1_L\rangle = \sqrt{\tfrac{2}{3}}|11\rangle|0\rangle - \sqrt{\tfrac{1}{3}}\left\{\tfrac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right)\right\}|1\rangle$$

Hi acts as σi on these *encoded qubits*

Total angular momentum operators on n qubits → Collective su(2) operators:

$$S_\alpha^n = \sum_{i=1}^{n} s_\alpha^{(i)} \quad \text{where } s_\alpha^{(i)} = I \otimes \ldots \otimes I \otimes \underbrace{s_\alpha}_{i\text{th qubit}} \otimes I \otimes \ldots \otimes I \qquad \left[S_\alpha^n, S_\beta^n\right] = i\varepsilon_{\alpha\beta\gamma} S_\gamma^n$$



Degeneracy in the creation of a spin-s particles out of n spin-1/2 particle = encoded qubits

Exchange commutes with total angular momentum operators, but mixes degeneracy: changes order of addition of angular momentum!

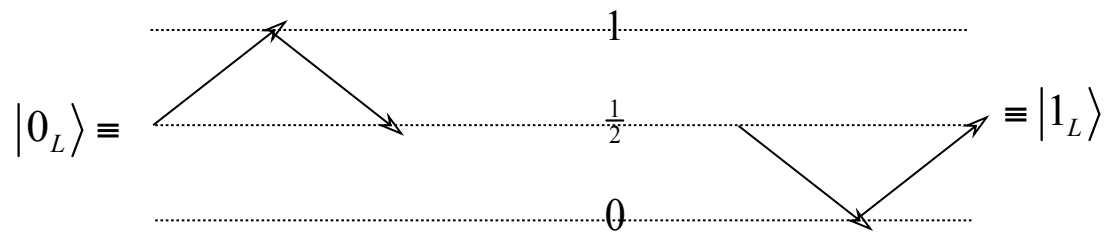
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n=3: the states are formed from combining a triplet with a spin-$\frac{1}{2}$ state or or by combining a singlet with a spin-$\frac{1}{2}$ state. Triplet is invarient under exchange and singlet flips sign under exchange. Thus

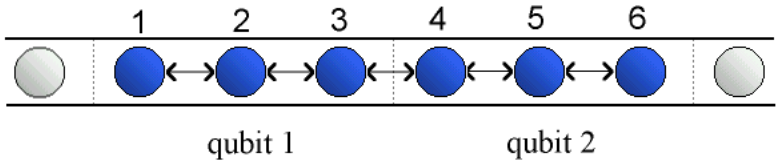
$E_{12} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ on the logical qubits.

more details tell us that

$E_{23} \rightarrow \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$ on the logical qubits.

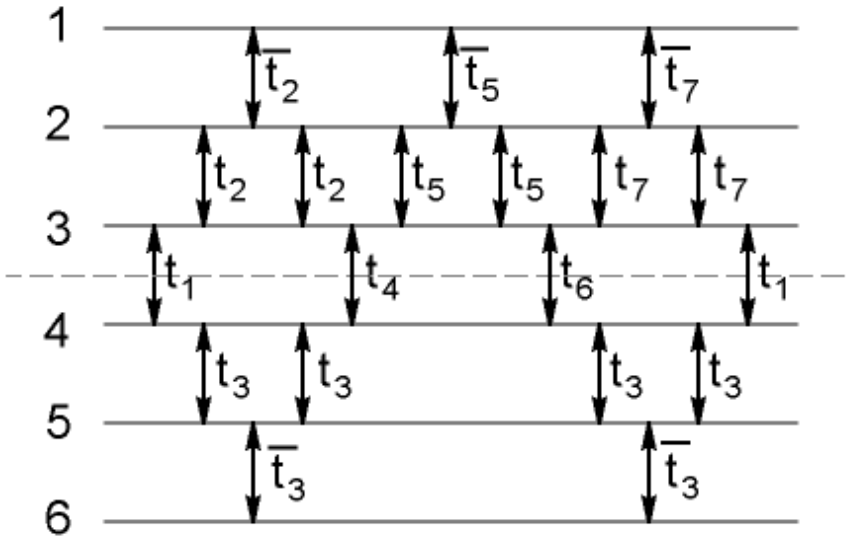


Explicit Gate Construction



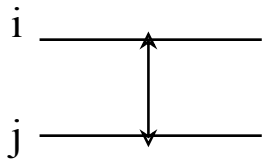
Nearest neighbor exchange coupling

How to couple encoded qubits?



t_1=0.410899(2) t_5=0.414720(10)
t_2=0.207110(20) t_6=0.147654(12)
t_3=0.2775258(12) t_7=0.813126(12)
t_4=0.640505(8) $\tan(\pi t)\tan(\pi \bar{t}) = -2$

exchange gate



$= \exp\left(iJE_{i,j}t\right)$

DiVincenzo, Bacon, Kempe, Burkard, Whaley, *Nature* **408**, 339 (2000)

<u>Tradeoffs</u>

factor of 3 in space

factor of ~ 10 time

What Do I Need To Quantum Compute?



Hey I found this new quantum system with these crazy interactions. Can I quantum compute?

- **"Encoded universality"** changes the rules for physical implementations of quantum computers
- **Exchange is universal with a bit of encoding**
- **Other interactions** besides exchange have similar properties (simple application of representation theory of Lie algebras: *take it to your local particle theorist for a quick answer!*)



Controlled-Not and single qubit gates are not the only way to build a quantum computer!



Acknowledgements

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